Summary
Forces show the existence of fields and how they interact with one another but here the force itself is discussed in more depth and then quantified. These forces also lead to the use of magnetic fields to induce electrical currents and the applications of this electromagnetic induction in motors, dynamos and transformers.

Underlying knowledge and understanding
This topic will predominantly be new content for learners with some understanding of D.C. motors. Learners will have looked at fields in the previous subtopic and now this knowledge will be built on to give learners the understanding of the application.

Common misconceptions
Learners find understanding the manner in which electric and magnetic fields interact to produce a force challenging. Learners commonly have difficulty with the right angles and three-dimensional requirements of Fleming's left-hand rule. Their ability to visualise this will impact how they deal with this concept. Learners find the action of a commutator difficult to apply in the D.C. motor. The application of changing direction of field in the transformer is found challenging by many learners and hence often leads to a superficial grasp of the working of the transformer.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
PM4.2i	**apply: force on a conductor (at right angles to a magnetic field) carrying a current:** **force (N) = magnetic flux density (T) × current (A) × length (m)**	**M1a, M1b, M1d, M2a, M3a, M3b, M3c, M3d**
PM4.2ii ☑	**apply:** $\dfrac{\textbf{potential difference across primary coil (V)}}{\textbf{potential difference across secondary coil (V)}} = \dfrac{\textbf{number of turns in primary coil}}{\textbf{number of turns in secondary coil}}$	**M1a, M1b, M1c, M1d, M2a, M3a, M3b, M3c, M3d**

Topic content			Opportunities to cover:		
Learning outcomes		**To include**	**Maths**	**Working scientifically**	**Practical suggestions**
P4.2a	**describe how a magnet and a current-carrying conductor exert a force on one another**			WS1.1b, WS1.1e, WS1.2a, WS1.3e	Demonstration of the jumping wire experiment.
P4.2b	**show that Fleming's left-hand rule represents the relative orientations of the force, the current and the magnetic field**				
P4.2c	**apply the equation that links the force on a conductor to the magnetic flux density, the current and the length of conductor to calculate the forces involved**		M1a, M1b, M1d, M2a, M3a, M3b, M3c, M3d		
P4.2d	**explain how the force exerted from a magnet and a current-carrying conductor is used to cause rotation in electric motors**	**an understanding of how electric motors work but knowledge of the structure of a motor is not expected**		WS1.1e, WS1.3e, WS2a	Construction of simple motors.
P4.2e ☑	**recall that a change in the magnetic field around a conductor can give rise to an induced potential difference across its ends, which could drive a current, generating a magnetic field that would oppose the original change**			WS1.1e, WS1.3e, WS2a	Examination of wind up radios or torches to investigate how dynamos work. Demonstration of induction using a strong magnet and a wire using a zero point galvanometer.
P4.2f ☑	**explain how this effect is used in an alternator to generate a.c., and in a dynamo to generate d.c.**			WS1.1a, WS1.1e, WS1.4a	Research the structure of dynamos and compare with DC motors.

P4.2g ☑	**explain how the effect of an alternating current in one circuit, in inducing a current in another, is used in transformers**				
P4.2h ☑	**explain how the ratio of the potential differences across the two coils in a transformer depends on the ratio of the numbers of turns in each**		M1c	WS1.1e, WS1.2a, WS1.2b, WS1.3a, WS1.3b, WS1.3e, WS1.3h, WS2a, WS2b	Building of a step-up and step-down transformer to investigate their effects.
P4.2i ☑	**apply the equations linking the potential differences and numbers of turns in the two coils of a transformer**		M1c, M3b, M3c		
P4.2j ☑	**explain the action of the microphone in converting the pressure variations in sound waves into variations in current in electrical circuits, and the reverse effect as used in loudspeakers and headphones**	**an understanding of how dynamic microphones work using electromagnetic induction**		WS1.1e, WS1.2a, WS1.3e, WS1.3h, WS2a, WS2b	Examination of the construction of a loudspeaker. Building of a loud speaker.